Response to Anonymous Referee #2

Manuscript Title: Climatology and trends of extreme precipitation in France: evaluation of an explicit-convection regional climate model

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Journal: Hydrology and Earth System Sciences (HESS)

We thank referee #2 for their constructive and detailed review of our manuscript. Below are our point-by-point responses to the comments, with line and figure numbers referring to the manuscript.

General Comments

Comment 1: Return Period Choice

There is a 10-year return period chosen for both daily and hourly data, even though 10-year return period for hourly is much more extreme compared to daily given the amount of hourly data in a 10-year period compared to daily. Has a lower return value been tested? Given the noisy result, and the shorter period of data a better justification for the high return period for hourly data should be given, and it is of interest to show a lower return period.

- **Response:** We thank the reviewer for this comment. We agree that estimating a 10-year return level (RL_{10}) at the hourly scale from a 33-year series (1990–2022) is more uncertain than at lower return periods, because each hourly annual maximum carries less information about the tail of the distribution.

As a sensitivity check, we computed relative trends in RL_2 and RL_5 using the same non-stationary GEV models and the same 1992–2022 trend window as for RL_{10} (Response Figure 1 and Response Figure 2). Quantitative comparisons (hydrological year, all stations with a selected GEV model) show that the spatial patterns are robust to the choice of return period, while trend magnitudes increase with T, as expected for higher quantiles:

Daily (1,583 stations; 704 with significant RL_{10} trends, 44.5%). Among significant stations, pairwise Pearson correlations of relative trends are $r = 0.99$ between RL_5 and RL_{10},

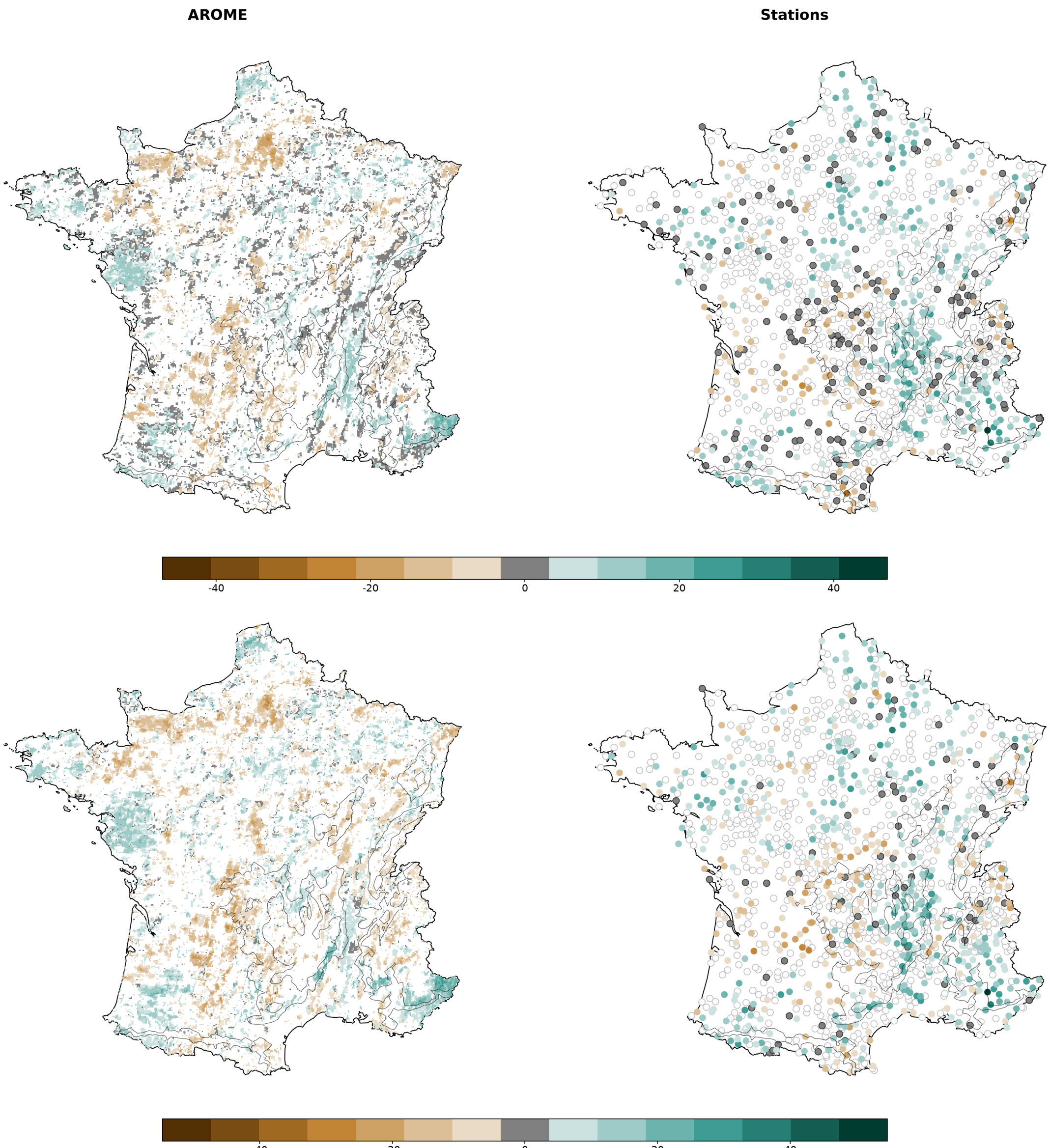
$r = 0.86$ between RL_2 and RL_{10}, and $r = 0.92$ between RL_2 and RL_5. Median absolute trends are similar across return periods (9.3%, 10.9%, and 11.3% for RL_2, RL_5, and RL_{10}, respectively). The same regional signals appear on all three maps—notably positive trends in the southern Alps and Mediterranean arc, and weaker or negative trends in parts of western France—with RL_5 virtually indistinguishable from RL_{10} at the national scale.

Hourly (574 stations; 193 with significant trends, 33.6%). Correlations remain high but are lower than for daily data, reflecting the shorter records and higher convective variability: $r = 0.98$ (RL_5 vs. RL_{10}), $r = 0.79$ (RL_2 vs. RL_{10}), and $r = 0.87$ (RL_2 vs. RL_5) among significant stations. Median absolute trends increase from 31% (RL_2) to 42% (RL_5) and 45% (RL_{10}), confirming that RL_{10} amplifies tail trends but does not create spurious regional structures: the same broad patterns are present at RL_2 and RL_5, with reduced amplitude and less spatial coherence, as expected for a lower quantile on a 33-year record.

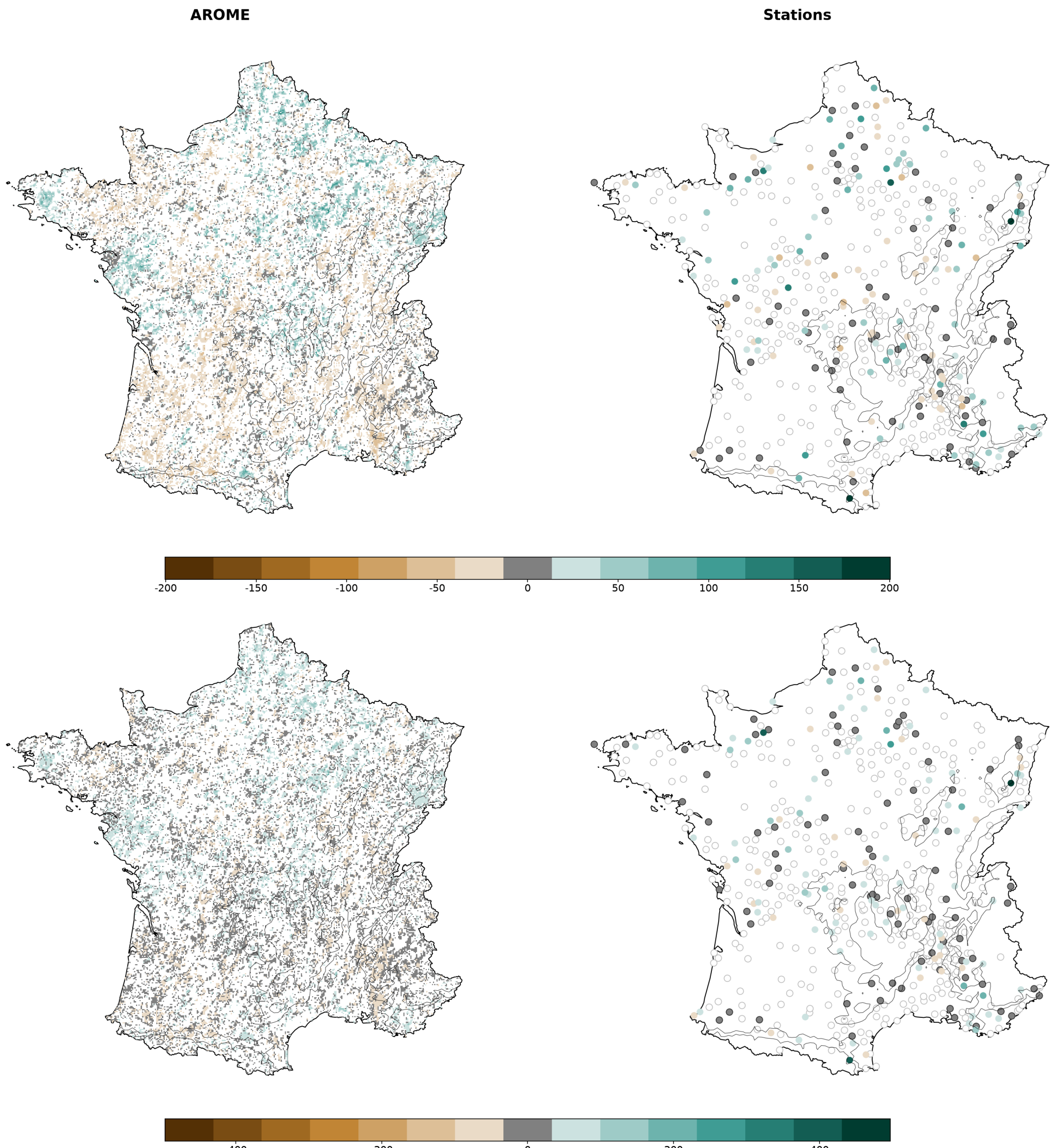
AROME shows the same behaviour: $r = 0.99$ between RL_5 and RL_{10} trends (daily and hourly, significant grid points) and $r = 0.83$–0.86 between RL_2 and RL_{10}, consistent with the station-based analysis.

We note that significance is assessed from the RL_{10} trend uncertainty (10% level); the same 704 daily and 193 hourly stations are therefore flagged across return periods. This is conservative for RL_2 and RL_5, which are less noisy than RL_{10} at the hourly scale. Together with the mean annual-maxima check in our response to Reviewer #1 ($r = 0.88$ for relative trends across 1,558 daily stations), these results indicate that the trend maps (Figures 6 and 7) are not driven by GEV fitting artefacts or by the choice of a high return period alone.

We retain RL_{10} as the primary metric because it is true extreme and because our daily (63-year) and hourly (33-year) records are sufficient for its estimation. In addition, using the same return period at both daily and hourly scales enables a direct and consistent comparison of the extreme trend characteristics between the two time resolutions. The results for RL_2 and RL_5 confirm the robustness of the detected spatial patterns. We present the lower-return-period maps here for the reviewer's assessment rather than as an additional manuscript appendix, to keep the main text focused.



Response Figure 1: Relative GEV trends (%) of RL_2 (first row) and RL_5 (second row) for daily precipitation (hydrological year): columns are AROME (left) and Météo-France stations (right), with a separate colour scale for each row. Display conventions: open circles for non-significant stations ; filled grey for significant near-zero trends; coloured markers for significant trends (uniform size); non-significant AROME grid cells not displayed.



Response Figure 2: Relative GEV trends (%) of RL_2 (first row) and RL_5 (second row) for hourly precipitation (hydrological year): columns are AROME (left) and Météo-France stations (right), with a separate colour scale for each row. Display conventions: open circles for non-significant stations ; filled grey for significant near-zero trends; coloured markers for significant trends (uniform size); non-significant AROME grid cells not displayed.

Manuscript modifications:

- In `Section 3.2.3 (Trend in return level)`, we have added the justification for using the 10-year return level as the main indicator, while noting the consistency with lower return periods:

> *"In this study, we focus on the 10-year return level (*RL_{10}*) as the primary metric for extreme precipitation. Our daily series (63 years) and hourly series (33 years) provide sufficient data for its robust estimation. As a sensitivity check, trends in* RL_2 *and* RL_5 *show the same regional patterns as* RL_{10} *(daily:* $r = 0.86$–0.99 *among significant stations; hourly:* $r = 0.79$–0.98*), confirming that the detected signals are robust and not driven by GEV fitting uncertainty."*

Comment 2: Introduction & Physical Processes

Introduction: First paragraph introduces a lot of different processes that are not necessarily connected. For instance, global warming leads to an increase in surface temperature, but it's the temperature increase in the atmosphere that increases the water holding capabilities, and the energy balance is also between the top of atmosphere and surface (and everything in between). A lot of the statements in Introduction need a reference for instance line 47-50 and should include more research articles in addition on the IPCC 2021 report.

- **Response:** We agree that the physical description in the first paragraph was somewhat compressed and mixed different thermodynamic and dynamic concepts. In the revised manuscript, we have restructured this paragraph to:

1. Clarify the physical link: surface warming leads to atmospheric warming, which increases the water-holding capacity of the troposphere according to the Clausius-Clapeyron relationship, while the actual condensation and precipitation are driven by vertical motion and adiabatic cooling of ascending air masses.
2. Add more academic references (e.g., Trenberth et al. 2003, O'Gorman and Muller 2010, Westra et al. 2014) alongside the IPCC 2021 report to ground these statements in the literature.

Manuscript modifications:

- In `Section 1 (Introduction and context)`, the first paragraph has been restructured. The main changes are:
  - Replace *"Due to buoyancy (Archimedes' principle), warm air surrounded by cooler air tends to rise. As warm air ascends in the atmosphere, it undergoes adiabatic cooling, leading to the condensation of water vapor and the formation of precipitation (Lin, 2022)."* with a clearer formulation that distinguishes thermodynamic and dynamic contributions.

- Replace *"However, under calm conditions, the central portion of the precipitation distribution does not fully capitalize on this excess moisture."* with *"However, for non-extreme or average precipitation events, [...]"* to clarify the distinction between mean and extreme precipitation.
- Add references: Trenberth et al. (2003), O'Gorman and Muller (2010), and Westra et al. (2014) alongside the existing IPCC (2021) and Clapeyron (1834) citations.
- Replace *"sensitivities"* (line 141) with *"temperature-scaling rates"* (as also requested by Reviewer #1).

Comment 3: Treatment of Aerosols and Solar Brightening

Figure 2 and Figure 3: How much of the temperature trend could be caused by aerosol changes due to anthropogenic pollution? How is aerosol treated in AROME? The results in these plots indicate that there is an increase in temperature after the peak pollution in the 1980s and that the trend in 10-year return level is increasing after this peak. Even though aerosols are not included or kept constant in AROME, this should be stated and included in the discussion and limitations.

- **Response:** We would like to clarify that aerosols are **not** kept constant in this AROME simulation: they are monthly evolving and vary from year to year, taken from the ALDERA reanalysis (which uses an interactive aerosol scheme to capture historical variations, including the peak pollution in the 1980s and subsequent solar dimming/brightening, as described by Nabat et al. 2020). Green House Gases also evolve annually. In the revised manuscript, we have made this clearer in Section 2.2 and Section 5.1.1 (Discussion).

Regarding the origin of the weaker simulated temperature trend, we do not attribute it to the ERA5 boundary forcing, which reproduces observed temperature trends well over France. More plausible internal model limitations include the model's sensitivity to radiative forcings, the absence of interactive vegetation (vegetation properties are prescribed and do not evolve with climate), and the use of a time-invariant land use/land cover map over the 63-year simulation period.

Manuscript modifications:

- In `Section 2.2 (CNRM-AROME model)` (line 183), the existing sentence *"Aerosols are monthly evolving and come from the ALDERA reanalysis, a CNRM-ALADIN simulation at 12 km driven by ERA5 with an interactive aerosol scheme (Nabat et al., 2020). Green House Gases are evolving annually."* already states this. We have expanded it to explicitly mention the peak pollution and dimming/brightening:

  > *"Aerosols are monthly evolving and vary from year to year, taken from the ALDERA reanalysis, a CNRM-ALADIN simulation at 12 km driven by ERA5*

with an interactive aerosol scheme that captures historical variations including the peak anthropogenic pollution in the 1980s and subsequent solar dimming/brightening (Nabat et al., 2020). Greenhouse gases also evolve annually."

- In `Section 5.1.1 (Temperature trends)`, after the existing paragraph on temperature trends (line 207: *"It should be noted that temperature trends [...] the magnitude of the trends we can detect."*), we have added:

  "The weaker simulated temperature trend suggests an internal dampening of the warming signal in France compared to observations. Since aerosols and greenhouse gases evolve realistically in this simulation (Section~2.2) and ERA5 reproduces observed temperature trends well, the underestimation likely stems from the model's internal sensitivity to radiative forcings, the absence of interactive vegetation, and/or the lack of temporal evolution of the land use/land cover map."

Comment 4: Visualization of Non-Significant Trends

For the maps with stations, setting non-significant trends to zeros makes it difficult to differentiate with stations that have close to zero or zero significant trends and stations with strong trends that are non-significant. The authors should consider not showing the stations at all or choosing another marker. The size of the marker also seems to vary with the trend, which can make it look like the stations with strong trends are more numerous than they are.

- **Response:** We agree with the reviewer. Representing non-significant trends as zero makes it difficult to distinguish between a significant trend of 0% and a non-significant trend of 50%. For the climatology maps (Figure 4), we only adopted uniform marker size and a clearly visible grey for zero bias (no significance coding). For the trend maps (Figures 6 and 7, and Appendix C), we adopted the following display conventions (also detailed in our response to Reviewer #1, Comment 7):

Type	Display
Non-significant (stations)	Open circle: white fill, light grey outline
Significant 0 (stations)	Filled grey disk, dark outline
Significant coloured (stations)	Trend colour on the map scale; **uniform marker size**
AROME grid	Non-significant grid points **not displayed**

This keeps the station network visible on hourly maps (where only about one third of stations are significant) while clearly separating non-significant locations from significant near-zero and coloured trends.

Manuscript modifications:

- In `Figures 6 and 7` (trend maps), we have changed the visualization according to the table above.
- The caption of Figures 6 has been updated. Original caption: *"Relative GEV trends over 1959–2022 in the 10-year return level of daily precipitation [...]."* Updated caption adds the display conventions (open circles for non-significant stations, filled grey for significant near-zero trends, coloured markers for significant trends, uniform size; non-significant AROME grid cells not displayed).
- The same changes have been applied to the captions of Figures 6 and 7 (hourly trends) and the monthly trend maps (Appendix C, Figure C1).

The Figure 3 caption has been updated for uniform marker size and zero-bias display; captions for Figures 5 and 6 and Appendix C, Figure C1 describe the significance conventions above. The revised maps are shown in our response to Reviewer #1 (Comment 7): Figure 3 (climatology) and Figures 5 and 6 (trends).

Minor Comments

- **Why choose the hydrological year? The study itself does not take into account snowpack, and in addition the convective extreme precipitation can come in summer that in this study is over the start of the hydrological year (JAS, where summer usually is considered as JJA). This should be justified more.**

  Response: We thank the reviewer for this comment, which highlights an important distinction between two levels of our analysis.

  For **annual maxima** (hydrological year, September to August), the September start is motivated by the need to keep the autumn Mediterranean convective season intact within a single block, as required for the independence assumption of annual maxima GEV modeling. This choice is standard in French climatology and hydrology when studying annual extremes.

  For **seasonal** analyses, the grouping (OND, JFM, AMJ, JAS) is a standard partition in French hydrology for extreme precipitation studies (e.g., Haruna et al., 2026). As shown by Berghald et al. (2025), this grouping is particularly suited for capturing extreme trend patterns compared to traditional meteorological seasons (e.g., SON, DJF).

  To avoid conflating these distinct convective regimes, we also provide **monthly** analyses (Figures and Appendix), which allow summer (June–August) and autumn (September–November) signals to be examined separately. We have clarified this distinction in the methodology section.

  Manuscript modifications: In `Section 3 (Methodology)`, after the definition of seasons and the hydrological year, we have updated the clarification as follows:

"The hydrological year (September to August) is used for annual maxima instead of the calendar year, which is standard in French climatology and hydrology. This choice is motivated by the fact that the most intense extreme precipitation events in France, particularly Mediterranean episodes (e.g., Cévenol events), occur during autumn (September–November). Using a calendar year would risk splitting these autumn events across two years, violating the block-independence assumption required for annual maxima GEV modeling. The seasonal partition (OND, JFM, AMJ, JAS) follows the conventional grouping of months used in France for precipitation extreme studies (Haruna et al., 2026). A previous study by Berghald et al. (2025) demonstrated that this grouping better captures trend patterns compared to the traditional seasonal definition (e.g., SON, DJF)."

- **It can be confusing in the text with GEV models and Arome model, could consider naming one different.**

  Response: We agree. In the revised text, we refer to CNRM-AROME as the "climate model", "simulation(s)", or "CP-RCM" to avoid any confusion, and have specifically replaced "AROME model" with "AROME simulations" in the results sections. We refer to the GEV models as "statistical models" or "statistical GEV distributions".

  Manuscript modifications: Throughout the manuscript, we have revised the terminology to prevent confusion between the climate model and the extreme value models. Specifically:

  - We refer to CNRM-AROME as the "CP-RCM", "climate model", or "simulation(s)", and have replaced "AROME model" with "AROME simulations" in the results sections.
  - We refer to GEV models (M_0 to M_3^*) as "statistical models" or "statistical GEV distributions".

- **Too little font in Figure 8**

  Response: Figure 8 has been moved to the Appendix, where it can be displayed at a larger scale and read more easily.

  Manuscript modifications: Figure 8 (monthly GEV trend synthesis) has been relocated to the Appendix to improve readability.